



Automated Washroom Light Using IR Sensors

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This project based on Arduino Uno and IR sensors is used to automatically switch on and switch off a washroom's light that works on AC mains. Whenever a

person enters the washroom, the light bulb (or tubelight) will automatically turn on. When the person leaves the bathroom, the light will turn off. This will save electricity charges due to the

light remaining on by mistake after use of the washroom. The author's prototype is shown in Fig. 1.

Circuit and working

Circuit diagram of the automated light is shown in Fig. 2. It is built around Arduino Uno (Board1), BC547 transistor (T1), infrared (IR) sensor modules (Module1 and Module2), and relay (RL1).

Infrared sensor. The IR sensors (FC-51) used here (refer Fig. 3) are commonly used in obstacle-detection projects. Two pairs of the IR sensor modules are used to detect a person's entry or exit from the washroom.

Each sensor has three pins, namely, Vcc, Vout, and GND. The Vcc pin is to be provided with 5V DC supply, which can be taken from 5V pin of the Arduino Uno. The ground pin (GND) of the sensor can be connected to GND pin of Arduino Uno. The Vout pins of IR Module1 and IR Module2 are the output pins connected to pins 8 and 7 of Arduino Uno, respectively.

The output of each sensor will be either 5V (high) or 0V (low) based on the detection. When a person is detected, Vout is low, otherwise it is high (about 5V). These logic levels of both the sensors are used to find out whether the person is entering or leaving the washroom.

5V relay. The 5V electromechanical relay (RL1) is used to switch on and switch off the light bulb. A typical sugar-cube type 5V relay has five terminals, two of which are used for energising the coil. The rest are common (COM), normally closed (NC), and normally open (NO) terminals.

The light bulb is connected between the NO and COM contacts of the relay. As is obvious, in normal condition, NO pin does not have any contact with the COM pin. When the relay coil

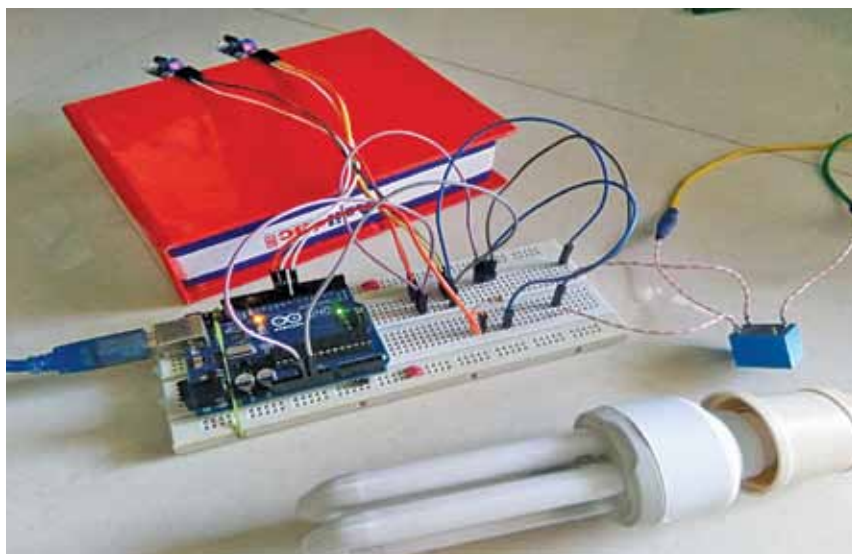


Fig. 1: Author's prototype

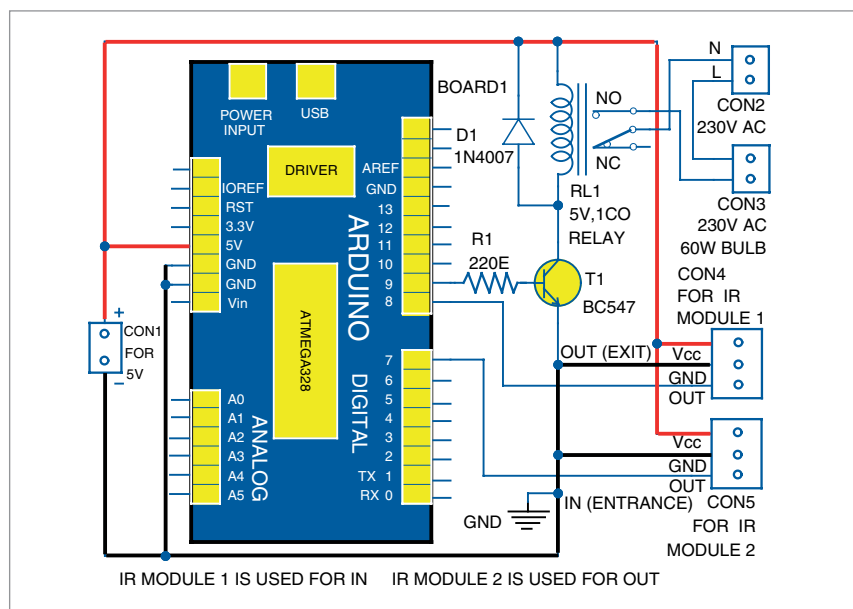


Fig. 2: Circuit diagram of automated washroom light